

correlation and Regression

1 Define co-efficient of correlation.

correlation co-efficient r is a statistical measure that quantifies the linear relationship between a pair of variables.

Define rank correlation.

The rank correlation is a non-parametric statistical procedure used for determining the bivariate correlation of two at least ordinal scaled attributes, whereby two ranking sequences are compared each other.

Why we need rank correlation?

The primary reason to use rank correlation is that it is robust outliers. It can also reveal non-linear relationships. We need rank correlation method when the rank-order data are available or when each variable can be ranked in some order.

2. Show that $R = 1 - \frac{6\sum d^2}{n(n^2-1)}$

Let x and y assume the same values $1, 2, \dots, n$

The correlation coefficients,

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n \sigma_x \sigma_y}$$

from which

$$n \sigma_x \sigma_y = \sum (x_i - \bar{x})(y_i - \bar{y}) \quad \text{--- (1)}$$

Since x and y assume the same values,

$$\bar{x} = \bar{y} \text{ and } \sigma_x^2 = \sigma_y^2$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{1+2+\dots+n}{n} = \frac{n+1}{2} = \bar{y}$$

$$\text{again, } n \sigma_x^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

$$= \frac{n(n+1)(2n+1)}{6} - \frac{\{n(n+1)\}^2}{4n}$$

$$= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)^2}{4}$$

$$= \frac{n(n+1)}{2} \left[\frac{2n+1}{3} - \frac{n+1}{2} \right]$$

$$= \frac{n(n+1)}{2} \times \frac{4n+2-3n-3}{6}$$

$$= \frac{n(n^2-1)}{12} \quad \text{--- (11)}$$

$$= n \sigma_y^2$$

Thus, $\sigma_x = \sigma_y$ — (11)

Since $\bar{x} = \bar{y}$, the difference d_i can be expressed

as, $d_i = (x_i - y_i) = (x_i - \bar{x}) - (y_i - \bar{y})$

$$\therefore \sum d_i^2 = \sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2 - 2 \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= n\sigma_x^2 + n\sigma_y^2 - 2nr\sigma_x\sigma_y \quad [\text{from i}]$$

$$= 2n\sigma_x^2(1-r) \quad [\text{from ii}]$$

$$\Rightarrow \frac{2n(n^2-1)(1-r)}{12} = \frac{6 \sum d_i^2}{12} \quad [\text{from iii}]$$

$$\Rightarrow 2n(n^2-1)(1-r) = 6 \sum d_i^2$$

$$\Rightarrow (1-r) = \frac{6 \sum d_i^2}{n(n^2-1)}$$

$$\therefore r = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

$$\therefore r = 1 - \frac{6 \sum d_i^2}{n(n^2-1)} \quad [\text{proved}]$$

Define scatter diagram:

A scatter diagram is a correlation chart that visually depicts the relationship between two variables.

3. Show that $-1 \leq r \leq 1$ or value of coefficient of correlation lies between -1 and 1.

We have,

The correlation coefficient,

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n \sigma_x \sigma_y}$$

$$\Rightarrow n r \sigma_x \sigma_y = \sum (x_i - \bar{x})(y_i - \bar{y}) \quad \text{--- (i)}$$

$$\text{where, } n \sigma_x^2 = \sum (x_i - \bar{x})^2 \quad \text{and} \quad \text{--- (ii)}$$

$$n \sigma_y^2 = \sum (y_i - \bar{y})^2 \quad \text{--- (iii)}$$

Let us consider the following:

$$\frac{x_i - \bar{x}}{\sigma_x} \pm \frac{y_i - \bar{y}}{\sigma_y} \quad \text{--- (iv)}$$

which, when squared is always positive,

$$\therefore \left(\frac{x_i - \bar{x}}{\sigma_x} \pm \frac{y_i - \bar{y}}{\sigma_y} \right)^2 \geq 0$$

$$\Rightarrow \frac{\sum (x_i - \bar{x})^2}{\sigma_x^2} + \frac{\sum (y_i - \bar{y})^2}{\sigma_y^2} \pm 2 \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y}$$

$$\Rightarrow \frac{n \sigma_x^2}{\sigma_x^2} + \frac{n \sigma_y^2}{\sigma_y^2} \pm 2nr \geq 0 \quad [\text{from (i), (ii) \& (iii)}]$$

$$\Rightarrow 2n \pm 2nr \geq 0$$

$$\Rightarrow (1 \pm r) \geq 0$$

$$\text{when } (1+r) \geq 0, \quad r \geq -1$$

$$\text{when } (1-r) \geq 0, \quad r \leq 1$$

$$\text{Thus, } -1 \leq r \leq 1 \quad [\text{proved}]$$

4. Prove that the coefficient of correlation is independent of origin and scale of measurement.

The correlation between x and y ,

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \quad \text{--- (1)}$$

Transform x and y to u and v .

$$u_i = \frac{x_i - a}{h} \quad \text{and} \quad v_i = \frac{y_i - b}{k}$$

so that $x_i = a + hu_i$ and $y_i = b + kv_i$ resulting in $\bar{x} = a + h\bar{u}$ and $\bar{y} = b + k\bar{v}$. Here a and b are the origin factors and h and k are the scale factors.

So, from (1),

$$r_{xy} = \frac{\sum (a + hu_i - a - h\bar{u})(b + kv_i - b - k\bar{v})}{\sqrt{\sum (a + hu_i - a - h\bar{u})^2} \sqrt{\sum (b + kv_i - b - k\bar{v})^2}} = \frac{\sum (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum (u_i - \bar{u})^2} \sqrt{\sum (v_i - \bar{v})^2}} = r_{uv}$$

which is expression that is independent of a, b, h and k . [proved]

6. 2020 Q4 5(d) [same - 2013]

Table for rank correlation coefficient

X	R _x	Y	R _y	d ² = (R _x - R _y) ²
50	9.5	11	10	0.25
50	9.5	12	8	2.25
55	8	14	7	1
60	5.5	16	5	0.25
65	2	12	9	49
65	2	19	2	0
65	2	20	1	4
60	5.5	15	6	0.25
60	5.5	17	4	0.25
60	5.5	18	3	6.25

$$\sum d^2 = 62.5$$

For sales (x),

65 repeated 3 times, so $m=3$; $\frac{3(3^2-1)}{12} = 2$

60 repeated 4 times, so $m=4$; $\frac{4(4^2-1)}{12} = 5$

50 repeated 2 times, so $m=2$; $\frac{2(2^2-1)}{12} = \frac{1}{2}$

$$\therefore R = 1 - \frac{6 \left(\sum di^2 + 2 + 5 + \frac{1}{2} \right)}{(n^2 - 1)n}$$

$$= 1 - \frac{6 \left(62.5 + 7 + \frac{1}{2} \right)}{40(40^2 - 1)}$$

$$= 0.575$$

Ans

2018 Q. 5(d) [same-2012]

NOTE - Rank না থাকলে ক্রম দিতে হবে। অরপক্ষ d^2 ক্রম দিতে হবে।

R_1	R_2	$d^2 = (R_1 - R_2)^2$
3	4	1
5	6	1
4	3	1
8	9	1
9	10	1
7	7	0
1	2	1
2	1	1
6	5	1
10	8	4

$$\sum d^2 = 12$$

$$\therefore R = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6 \times 12}{10(10^2 - 1)} = 0.927 \quad \text{(Ans)}$$

2021 Q. 5(c). [same-2019]

We know, Karl Pearson's coefficient of correlation,

$$r_{xy} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$\sigma_y = \frac{\text{cov}(x, y)}{r_{xy} \sigma_x}$$

$$= \frac{8.2}{0.75 \times 3}$$

$$= 3.64$$

$$\sigma_y^2 = (3.64)^2 = 13.28$$

Given,

$$r_{xy} = 0.75$$

$$\text{cov}(x, y) = 8.2$$

$$\sigma_x = 3$$

$$\sigma_y^2 = ?$$

2019. 5 (c)

We know,

Correlation of co-efficient,

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}} \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}}$$

$$= \frac{364 - \frac{56 \times 40}{8}}{\sqrt{524 - \frac{56^2}{8}} \sqrt{256 - \frac{40^2}{8}}}$$

$$= 0.98 \quad (\text{Ans})$$

Given, $\sum x = 56$

$\sum y = 40$, $\sum x^2 = 524$

$\sum y^2 = 256$

$\sum xy = 364$

$n = 8$

2021 Q4. 5 (b)

We know,

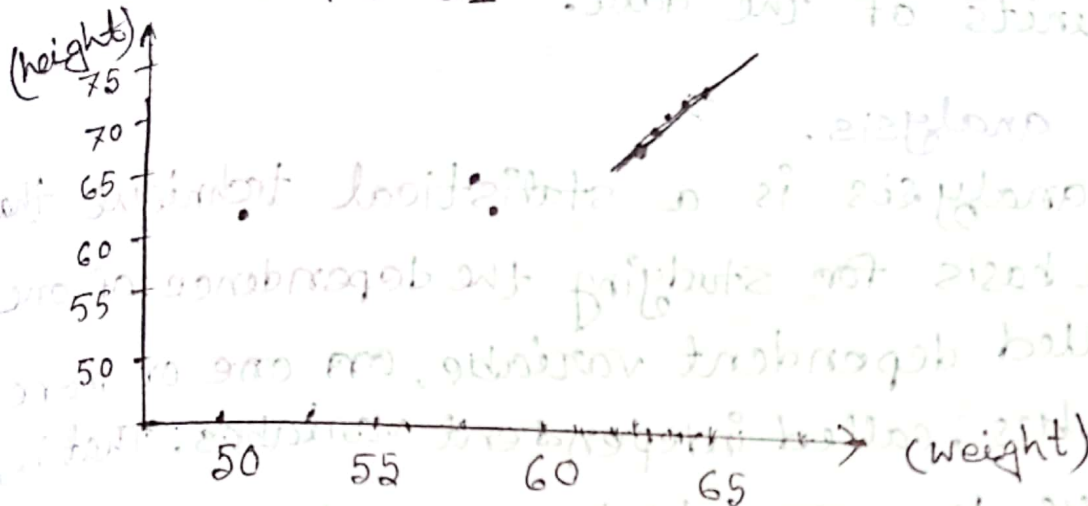
$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

height, x	weight, y	$x - \bar{x}$	$(x - \bar{x})^2$	$(y - \bar{y})$	$(x - \bar{x})(y - \bar{y})$	$(y - \bar{y})^2$
62	50	-5.25	27.56	-10.5	55.125	110.25
72	65	4.75	22.56	4.5	21.375	20.25
68	64	0.75	0.56	3.5	2.625	12.25
64	56	-3.25	10.56	-4.5	14.625	20.25
70	62	2.75	7.56	1.5	4.125	2.25
66	63	-1.25	1.56	2.5	-3.125	6.25
64	60	-3.25	10.56	-0.5	1.625	0.25
72	64	4.75	22.56	3.5	16.625	12.25
$\sum x = 538$	$\sum y = 484$	$\sum = 0$	$\sum = 103.48$		$\sum = 113$	$\sum = 184$

$$\therefore \bar{x} = \frac{538}{8} = 67.25$$

$$\text{and, } \bar{y} = \frac{484}{8} = 60.5$$

$$\therefore r = \frac{113}{\sqrt{103.48} \sqrt{184}} = +0.81$$



from the diagram, it can be said that the correlation is positive.

2021 Q. 6(c)

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$y - \bar{y}$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
10	-0.8	0.64	8	0	0	0
12	1.2	1.44	7	-1	1	-1.2
21	10.2	104.04	17	9	81	91.8
4	-6.8	46.24	2	-6	36	-40.8
7	-3.8	14.44	6	-2	4	-7.6
$\Sigma x = 54$		$\Sigma = 166.8$	$\Sigma y = 40$		$\Sigma = 122$	$\Sigma = 139$

$$\therefore \bar{x} = \frac{54}{5} = 10.8$$

$$\bar{y} = \frac{40}{5} = 8$$

$$\therefore r = \frac{\Sigma (x - \bar{x})(y - \bar{y})}{\sqrt{\Sigma (x - \bar{x})^2} \sqrt{\Sigma (y - \bar{y})^2}} = \frac{139}{\sqrt{166.8} \sqrt{122}}$$

$$= 0.97$$

Do

7. Define Regression.

Regression is the measure of the average relationship between two or more variables in terms of the original units of the data.

Regression analysis.

Regression analysis is a statistical technique that serves as a basis for studying the dependence of one variable, called dependent variable, on one or more other variables, called independent variables. That is, the technique through which regression equation is developed and thus estimates the value of dependent variable is called regression analysis.

Simple regression.

The measure of the average relationship between selected values of x and observed values of y in terms of the original units of the data is called simple regression.

multiple regression.

The measure of the average relationship between two or more other variables in terms of the original unit is called multiple regression.

co-efficient of regression

The co-efficient of regression gives the value by which one variable increases for a unit increase in the other variable.

The regression co-efficient are defined as follows,

Regression co-efficient of X on Y, b_{xy}

$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$$

Regression co-efficient of Y on X, b_{yx}

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

[NOTE: co-efficient of correlation, $r = \sqrt{b_{xy} \cdot b_{yx}}$]

Difference between correlation and regression:

Correlation	Regression
Correlation measures the degree of relationship between two or more variables.	measures the average relationship between two or more variables in terms of the original units of the data
Both the variables X and Y are random variables.	X is a random variable and Y is fixed.
Correlation co-efficient are symmetric.	are asymmetric
Correlation does not indicate the cause and effect relationship between variables.	Regression indicates.
It has limited application as it is confined only to linear relationship between variables.	It has wider application as it studies linear and non-linear relationship between variables.

8. How do you estimate the regression lines?

The regression line passes through the mean values of y and x . The mean of y is equal to the $\bar{y}$.

To estimate the regression lines, the following values need to be calculated:

$$a = \bar{y} - b\bar{x} \quad \text{and} \quad b = r \frac{\sigma_y}{\sigma_x}$$

The easiest way of calculating them is by using a table.

start off by working out the mean of the independent and dependent variables, $\bar{x}$ and $\bar{y}$.

we have, $\hat{y} = a + bx$

$$= (\bar{y} - b\bar{x}) + bx$$

$$= \bar{y} + b(x - \bar{x})$$

~~This is how we can estimate the regression lines.~~

for x on y the following values need to be calculated: $a = \bar{x} - b\bar{y}$ and $b = r \frac{\sigma_x}{\sigma_y}$.

$$\hat{x} = a + b\hat{y}$$

$$= \bar{x} - b\bar{y} + b\hat{y}$$

$$= \bar{x} + b(\hat{y} - \bar{y})$$

This is how the regression line is estimated.

2020 Q. 6(d)

[Same - 2019 - byx regression]

We know, For Y on X,

regression equation is $y = a + b_{yx}x$

and $b_{yx} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$, $a = \bar{y} - b_{yx}\bar{x}$

X	Y	X ²	Y ²	XY
30	28	900	784	840
45	40	2025	1600	1800
32	27	1024	729	864
28	24	784	576	672
27	26	729	676	702
24	18	576	324	432
$\Sigma = 186$	$\Sigma = 163$	$\Sigma = 6038$	$\Sigma = 4689$	$\Sigma xy = 5310$

here, $\bar{x} = \frac{\sum x}{n} = \frac{186}{6} = 31$, $\bar{y} = \frac{\sum y}{n} = \frac{163}{6} = 27.16$

$b_{yx} = \frac{5310 - \frac{186 \times 163}{6}}{6038 - \frac{(186)^2}{6}}$

$= \frac{5310 - 5083}{6038 - 5892} = \frac{227}{146} = 1.5548$ (ii)

$\therefore a = \bar{y} - b_{yx}\bar{x} = 27.16 - 1.5548 \times 31 = -2.13$

$\therefore$ The regression equation is, $y = -2.13 + 1.5548x$

Ans

* 2012 - Same [2008]

X	Y	X ²	Y ²	XY
22	18	484		396
23	20	529		460
23	21	529		483
24	20	576		480
26	21	676		546
27	22	729		594
27	23	729		621
28	24	784		672
30	25	900		750
30	26	900		780
$\Sigma X = 260$	$\Sigma Y = 220$	$\Sigma X^2 = 6836$		$\Sigma XY = 5782$

(i) Here, $\bar{X} = \frac{\Sigma X}{n} = \frac{260}{10} = 26$, $\bar{Y} = \frac{220}{10} = 22$

$$b_{yx} = \frac{\Sigma XY - \frac{\Sigma X \Sigma Y}{n}}{\Sigma X^2 - \frac{(\Sigma X)^2}{n}} = \frac{5782 - \frac{260 \times 220}{10}}{6836 - \frac{(260)^2}{10}} = 0.82$$

$$\therefore a = \bar{Y} - b_{yx} \bar{X} = 22 - 0.82 \times 26 = 0.78$$

$\therefore$ Regression equation, $Y = 0.78 + 0.82X$

(ii) Given, $Y = 30$

we have,

$$Y = 0.78 + 0.82X$$

$$\Rightarrow 30 - 0.78 = 0.82X$$

$$\Rightarrow X = \frac{29.22}{0.82}$$

$$= 35.63$$

$\therefore$ Husband's age is 35 years (approx.)

* 2021 Q9, 6(d) [same - 2018]

(6)2 - 9102

(i) Given, $12x - 15y + 99 = 0$ — (i)
 $60x - 27y - 321 = 0$ — (ii)

(i) $\times 5$ — (i) $\Rightarrow 60x - 75y + 495 = 0$

$60x - 75y + 495 = 0$

$60x - 27y - 321 = 0$

$\begin{array}{r} 60x - 75y + 495 = 0 \\ 60x - 27y - 321 = 0 \\ \hline -48y + 816 = 0 \end{array}$

from (i),

$12x - 15 \times 17 + 99 = 0$

$\therefore x = 13$, that is $\bar{x} = 13$

(2)

(ii) Given, $12x - 15y + 99 = 0$

$15y = 12x + 99$

$y = 0.8x + 6.6$

$\therefore b_{yx} = 0.8$

and, $60x - 27y - 321 = 0$

$x = \frac{27}{60}y + \frac{321}{60} = 0.45y + 5.35$

$\therefore b_{xy} = 0.45$

$\therefore$ coefficient of correlation, $r = \sqrt{b_{xy} \times b_{yx}}$

$= \sqrt{0.45 \times 0.8}$
 $= 0.6$ (2)

(iii) Given, $\sigma_x = 6$

$\therefore b_{xy} = r \frac{\sigma_x}{\sigma_y}$

$\therefore \sigma_y = \frac{r \sigma_x}{b_{xy}} = \frac{0.6 \times 6}{0.45} = 8$

$\therefore$ Variance $\sigma_y^2 = 8^2 = 64$

(2)

2019 - 6(c)

(i) Given, $\bar{x} = 35$, $\bar{y} = 82$, $r = 0.61$

$$\sigma_x = 0.03, \quad \sigma_y = 17.16$$

for regression x on y ,

$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = 0.61 \times \frac{0.03}{17.16} = 0.00107$$

$$a = \bar{x} - b_{xy} \bar{y} = 35 - 0.00107 \times 82 = 34.9114$$

$$\therefore x = a + b_{xy} y$$
$$= 34.9114 + 0.00107 y$$

for regression y on x ,

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = 0.61 \times \frac{17.16}{0.03} = 354.12$$

$$a = \bar{y} - b_{yx} \bar{x} = 82 - 354.12 \times 35 = -12498.2$$

$$\therefore y = a + b_{yx} x = -12498.2 + 354.12 x$$

(ii) Given, $x = 40$

$$\therefore y = -12498.2 + 354.12 \times 40$$

$$= 8776.8$$

bid Given, $y = 90$.

$$\therefore x = 34.9114 + 0.00107 \times 90$$
$$= 34.9114 + 0.000963$$
$$= 34.91236$$